

Simulated Data — sparsity (holes) in pseudotime

Replaces the `latent_t` row only. Everything downstream ($s_b, z_i, y_i = z_i W + s_b$, counts) is unchanged.

Throughout both slides, m and ρ are the *same* two numbers at every occurrence — there is only one of each.

1. Cut $[0, 1]$ into $2m + 1$ chunks that alternate dense / hole, starting and ending dense.

$$w_k = \begin{cases} 1 & k \text{ even} \\ \rho & k \text{ odd} \end{cases} \quad p_k = \frac{w_k}{Z}$$

$$Z = \sum_{k=0}^{2m} w_k = \underbrace{(m+1)}_{\# \text{ dense}} \cdot 1 + \underbrace{m}_{\# \text{ holes}} \cdot \rho$$

w_k = unnormalized weight (line 195).

Z = their sum; dividing by it (line 196) makes $\sum_k p_k = 1$.

p_k = probability one cell lands in chunk k = `chunk_prob[k]`.

the $m = 3, \rho = 0.05$ instance							
k	0	1	2	3	4	5	6
chunk	dense	hole	dense	hole	dense	hole	dense
weight w_k	1	ρ	1	ρ	1	ρ	1
=	1	0.05	1	0.05	1	0.05	1
$p_k = w_k/Z$	.2410	.0120	.2410	.0120	.2410	.0120	.2410

$$Z = 4 \cdot 1 + 3 \cdot 0.05 = 4.15$$

“ k even” and “dense” are the *same* condition, not two.

2. Draw one cell's pseudotime with two draws: *which* chunk, then *where inside* it.

$$k_i \sim \text{Cat}(p_0, \dots, p_{2m})$$

which chunk. chunk, line 199

$$u_i \sim \mathcal{U}(0, 1)$$

position *within* that chunk — not pseudotime.

latent_t_chunk, line 201

$$t_i = \frac{k_i + u_i}{2m + 1}$$

pseudotime, latent_t, lines 202–203

Cat = categorical (weighted die) over $\{0, \dots, 2m\}$; in code, `rng.choice(..., p=chunk_prob)`.

a real cell (cell_seed=4, cell 24):

$$t_i = \frac{k_i = 3 \text{ (a hole)}, \quad u_i = 0.9219}{3 + 0.9219} = \frac{3.9219}{7} = 0.5603$$

chunk 3 spans $[0.4286, 0.5714]$, and 0.5603 sits 92.19% of the way across it — which is what u_i means.

Derivation: lines 202–203 are $u \cdot \text{chunk_size}[k] + \text{chunk_offset}[k]$, then $/\text{total_size}$. At $h = 1$ every chunk_size is 1, $\text{chunk_offset}[k] = k$, $\text{total_size} = 2m+1$. Substitute. Not a standard formula — just those two lines.

Sparsity (holes) — the equivalent density

Marginalizing k_i out of slide 1 gives a two-level step density on $[0, 1]$. Same distribution, one line.
Same m and ρ as slide 1.

$$t_i \sim f_T, \quad f_T(t) = \frac{2m+1}{Z} \begin{cases} 1 & t \notin \mathcal{H} \\ \rho & t \in \mathcal{H} \end{cases}$$

$$\mathcal{H} = \bigcup_{j=1}^m \left[\frac{2j-1}{2m+1}, \frac{2j}{2m+1} \right)$$

j indexes the holes, 1 to m . It is a counter, not a parameter.

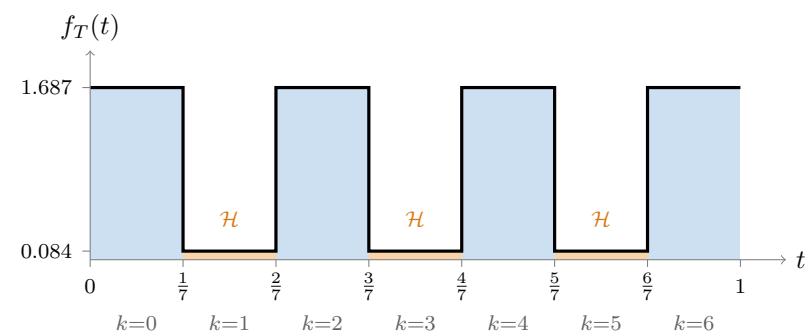
m = `n.holes`: how many low steps. ρ = `hole.density`: how *low* they are.

h = `hole.size`: how *wide* they are. $h = 1$ always, so all steps are equally wide and h drops out above.

General h : $S = (m+1) + mh$, $Z = (m+1) + mh\rho$, $f_T = \frac{S}{Z} \rho^{1[t \in \mathcal{H}]}$.

$m = 0$: one chunk, $Z = 1$, $k_i \equiv 0$, so $t_i = u_i \sim \mathcal{U}(0, 1)$ — the control, unchanged.

$\rho = 1$: $Z = 2m+1$, so $f_T \equiv 1$ — also $\mathcal{U}(0, 1)$. A hole at full density is not a hole.



Gaps in pseudotime *support* — those cells are never drawn. Nothing here zeroes x ; `adata.X` stays (2500, 200).